FINITE-TYPE ALGEBRAIC SPACES FORM A STACK

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 752-758 of sources/stacks-examples-stacks/pinned-original-source.tex; proof span is lines 760-823, the complete author proof (62 body lines).

Editorial note. Navigation stubs: 7 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 1 of queue_examples-stacks.json (pb=2629, stmt 752-758, proof 760-823, halfA).

Editorial note. stacks-examples-stacks wave-1 (exs1): fresh adjudication accept (2026-10-14); pool per Chief 09:4x authorization (idle-line throughput over kun's backup note, SMC precedent); examples-stacks.tex 27 labels/20 with proof judged 9 accept / 11 reject; claim_key v2 stacks:examples-stacks.tex:<label>.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0ADV: sources/stacks-examples-stacks/pinned-original-source.tex, source lines 823–823. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 04UA: sources/stacks-examples-stacks/pinned-original-source.tex, source lines 675–679. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 04U9: sources/stacks-examples-stacks/pinned-original-source.tex, source lines 633–637. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 041T: sources/stacks-examples-stacks/pinned-original-source.tex, source lines 823–823. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 03XH: sources/stacks-examples-stacks/pinned-original-source.tex, source lines 823–823. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:examples-stacks.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 026F: sources/stacks-examples-stacks/pinned-original-source.tex, source lines 823–823. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 042V: sources/stacks-examples-stacks/pinned-original-source.tex, source lines 823–823. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *The functor $p_{ft} : \mathcal{S}paces_{ft} \to (Sch/S)_{fppf}$ satisfies the conditions (1), (2) and (3) of Stacks, Definition 026F.*

Proof. We are going to write this out in ridiculous detail (which may make it hard to see what is going on).

We have seen in Lemma 04U9 that a morphism $(f, g) : X/U \to Y/V$ of $\mathcal{S}paces$ is strongly cartesian if the induced morphism $f : X \to U \times_V Y$ is an isomorphism. Note that if $Y \to V$ is of finite type then also $U \times_V Y \to U$ is of finite type, see Morphisms of Spaces, Lemma 03XH. So if $(f, g) : X/U \to Y/V$ of $\mathcal{S}paces$ is strongly cartesian in $\mathcal{S}paces$ and Y/V is an object of $\mathcal{S}paces_{ft}$ then automatically also X/U is an object of $\mathcal{S}paces_{ft}$, and of course (f, g) is also strongly cartesian in $\mathcal{S}paces_{ft}$. In this way we conclude that $\mathcal{S}paces_{ft}$ is a fibred category over $(Sch/S)_{fppf}$. This proves (1).

The argument above also shows that the inclusion functor $\mathcal{S}paces_{ft} \to \mathcal{S}paces$ transforms strongly cartesian morphisms into strongly cartesian morphisms. In other words $\mathcal{S}paces_{ft} \to \mathcal{S}paces$ is a 1-morphism of fibred categories over $(Sch/S)_{fppf}$.

Let $U \in \mathrm{Ob}((Sch/S)_{fppf})$. Let X, Y be algebraic spaces of finite type over U. By Stacks, Lemma 042V we obtain a map of presheaves

$$Mor_{\mathcal{S}paces_{ft}}(X, Y) \longrightarrow Mor_{\mathcal{S}paces}(X, Y)$$

which is an isomorphism as $\mathcal{S}paces_{ft}$ is a full subcategory of $\mathcal{S}paces$. Hence the left hand side is a sheaf, because in Lemma 04UA we showed the right hand side is a sheaf. This proves (2).

To prove condition (3) of Stacks, Definition 026F we have to show the following: Given

(1) a covering $\{U_i \to U\}_{i \in I}$ of $(Sch/S)_{fppf}$,
(2) for each $i \in I$ an algebraic space X_i of finite type over U_i, and
(3) for each $i, j \in I$ an isomorphism $\varphi_{ij} : X_i \times_U U_j \to U_i \times_U X_j$ of algebraic spaces over $U_i \times_U U_j$ satisfying the cocycle condition over $U_i \times_U U_j \times_U U_k$,

there exists an algebraic space X of finite type over U and isomorphisms $X_{U_i} \cong X_i$ over U_i recovering the isomorphisms φ_{ij}. This follows from Bootstrap, Lemma 0ADV part (2). By Descent on Spaces, Lemma 041T we see that $X \to U$ is of finite type which concludes the proof. □